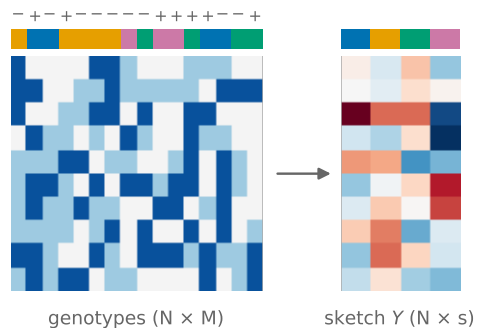


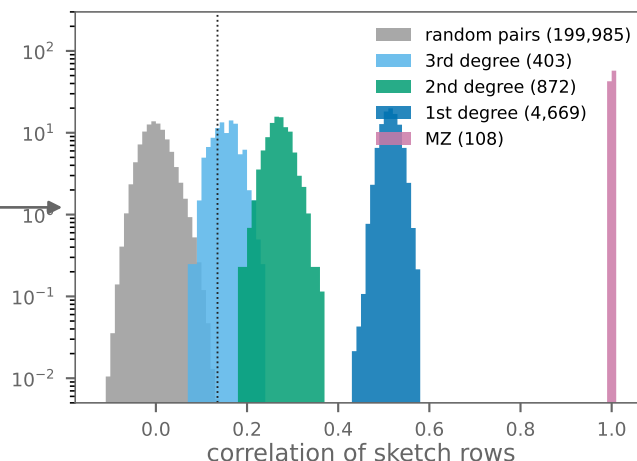
Very large N (N > 20,000): sketch search, then dwg

--robust (auto); --king-search auto

1 count sketch in pass 1: SNP j to column $h(j)$ with sign $\pm$



2 relatives stand out ($N = 20,000$, $s = 2048$)
unrelated: $\text{sd } 0.030 \approx 1/\sqrt{s}$



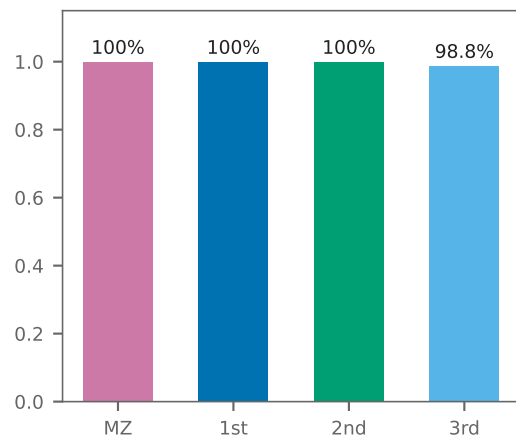
3 nearest neighbours

top $m = 5$ per person:
blocked float products,
cost N^2s , not N^2M

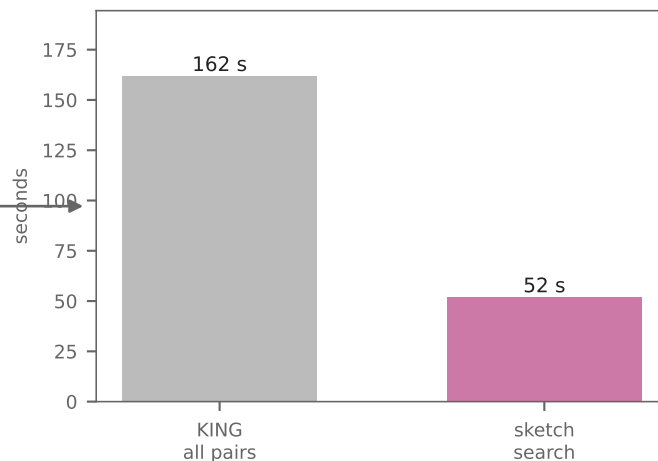
all m are relatives?
search $2m$, $4m$, ...

exact KING-robust on
 ~ 3.5 candidates / person

4 recall vs all-pairs KING
($N = 20,000$; same pairs at 100k)



5 finding the relatives
 $N = 100,000$, 16 threads



6 then as for large N

dwg fit, residual-sketch
evalAdmix + k_0 screen,
whitening (operator engine)

final relatedness:
 ϕ , k_0 , k_1 , k_2 (1 pass)

$N = 100k$: dwg 202 s
(detect-white 118 s;
standard PCAone 148 s)

Sketch: $N \times M$ (columns and signs from a hash of the SNP index; in-core = out-of-core). Search: N^2s , independent of M .
Relatives: correlation about 2ϕ ; unrelated: $0 \pm 1/\sqrt{s}$ (dotted: expected maximum over N). 1-bit signs and LSH bucketing do not work.